

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

TABLEAU DASHBOARD PROJECT

Sales Performance Analysis Dashboard

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Class / Batch	TYIT/1
Subject	Data Visualization with Power BI and Tableau
Academic Year	2026

Submitted to: SUMITKUMAR TRIPATHI

1. Project Title:

Blood Donation Analysis Dashboard using Tableau

2. Objective:

The main objective of this project is to create an interactive dashboard using Tableau for analyzing blood donation data. The dashboard converts raw donor data into meaningful visual information and helps users understand blood group distribution, donor demographics, donation frequency, donor eligibility, regular donor patterns, regional distribution, and donation trends over time.

3. Project Description:

This project uses Tableau to develop an interactive **Blood Donation Analysis Dashboard**. The dashboard contains key performance indicators (KPIs), charts, filters, and visualizations that allow users to explore the donor dataset. Users can analyze blood donation patterns based on blood type, age, gender, region, eligibility status, donation frequency, and donor status. The project demonstrates how data visualization can be used to identify patterns in blood donation data and communicate meaningful information clearly.

4. Dataset Details:

Parameter	Details
Dataset Name	Blood Donation Registry Dataset
Source	Hugging Face
Number of Records	30,000
Important Fields	Donor ID, Blood Type, Age, Sex, Region, Donation Count, Eligibility Status, Regular Donor Status, Last Donation Date

5. Tools & Technologies Used:

- **Tableau Desktop / Tableau Public**
- **Microsoft Excel / CSV** for dataset
- **Data Visualization and Dashboarding**

6. Tableau Worksheets / Charts:

Visualization	Purpose
KPI Cards	Total Donors, Total Donations, Eligible Donors and Regular Donors
Line Chart	Donation trend over time based on last donation date
Bar Chart	Number of donors by Blood Type
Bar Chart	Donors by Age Group
Map / Bar Chart	Donor distribution by Region / Country
Donor Status Chart	Eligible vs Deferred Donors
Deferral Reasons Chart	Distribution of donor deferral reasons

Donor Frequency Chart	Donors grouped by donation frequency
Interactive Filters	Blood Type, Region, Sex, Eligibility Status, Regular Donor Status and Year

The screenshot shows a data visualization tool interface. On the left, there's a sidebar with 'Connections' and 'Files'. The main area displays a table named 'blood_donation_registry_ml_ready.csv' with 27 fields and 30,000 rows. The table has columns: Donor Id, Age, Sex, Country Code, Region, and Blood Type. Below the table, there's a 'Fields' section with a table listing the fields and their physical tables.

Type	Field Name	Physical Table	Rem...
#	Donor Id	blood_donatio...	donor...
#	Age	blood_donatio...	age
Abc	Sex	blood_donatio...	sex
🌐	Country Code	blood_donatio...	countr...
Abc	Region	blood_donatio...	region
Abc	Blood Type	blood_donatio...	blood

Worksheet 1: KPI Cards

Visualization: KPI Cards

Purpose:

To display important blood donation metrics in a concise format and provide a quick overview of the donor dataset.

Fields Used:

- **Total Donors** — COUNTD(Donor Id)
- **Donation Count Last 12M** — SUM(Donation Count Last 12M)
- **Eligible To Donate** — SUM(Eligible To Donate)
- **Is Regular Donor** — SUM(Is Regular Donor)

Insights

1. The dataset contains **30,000 unique donors**, providing a substantial donor base for analysis.
2. **24,915 donors are eligible to donate**, while **5,685 are identified as regular donors**.
3. The recorded **donation count in the last 12 months is 22,958**, providing a measure of recent donation activity.

The KPI overview shows a donor base of 30,000 individuals, with 24,915 eligible donors and 5,685 regular donors. The dataset records 22,958 donations in the last 12 months.

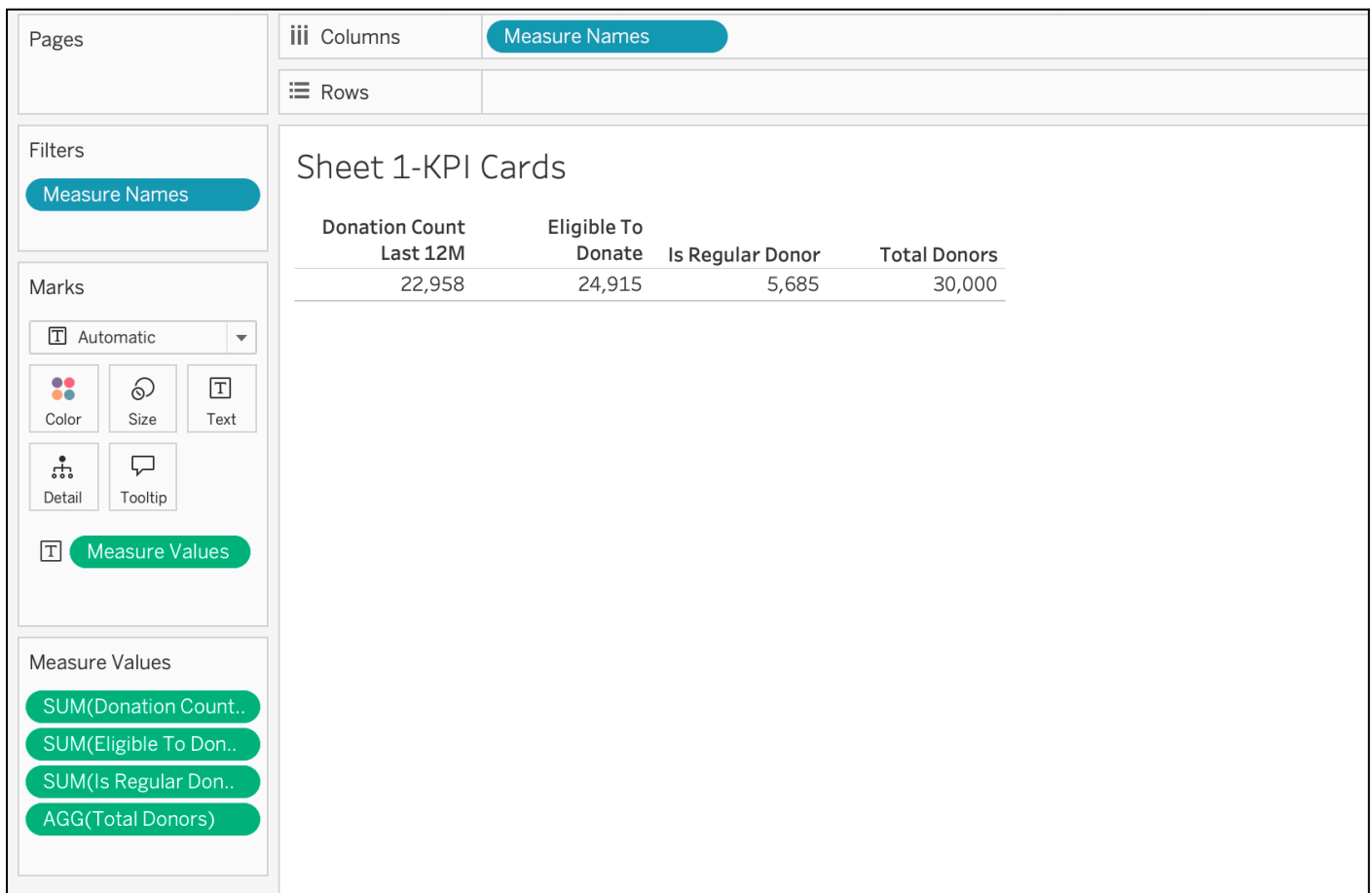


Figure 1: KPI Cards displaying key blood donation metrics including total donors, recent donation count, eligible donors, and regular donors.

Worksheet 2: Blood Type Analysis

Visualization: Bar Chart

Purpose:

To compare the number of donors across different blood types and understand the distribution of donors among the available blood groups.

Fields Used:

- **Blood Type** — Columns
- **Donor Id** — Rows
- **Aggregation:** Count Distinct (CNTD(Donor Id))
- **Mark Type:** Bar
- **Mark Labels:** Enabled

Insights

1. **O+** has the highest number of donors, with **12,649** donors.
2. **AB-** has the lowest number of donors, with **355** donors.
3. The dataset shows a much larger representation of **positive blood types** compared with negative blood types.

O+ represents the largest donor group with 12,649 donors, whereas AB- represents the smallest group with 355 donors. The distribution shows substantially higher donor counts for positive blood types than negative blood types.

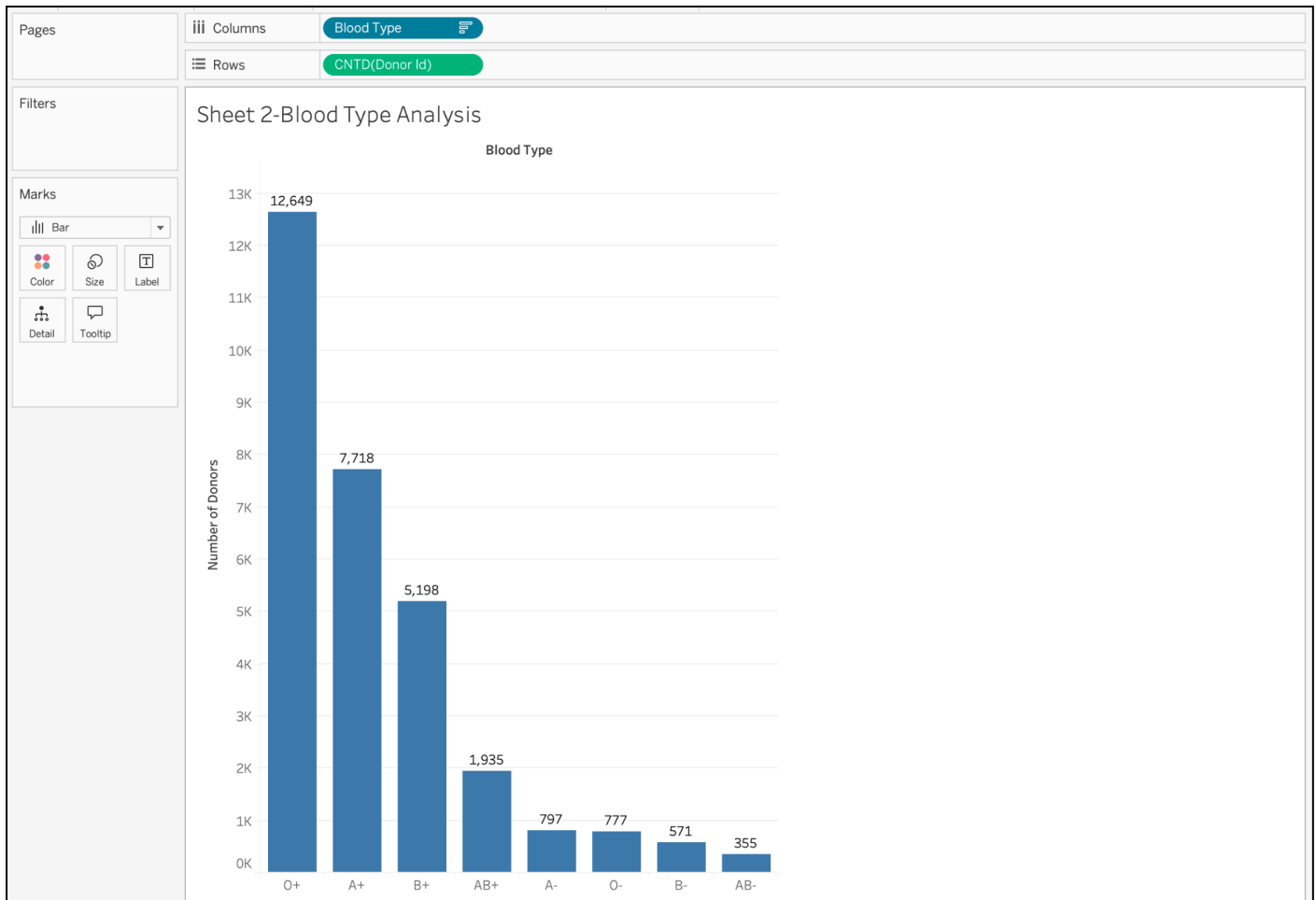


Figure 2: Blood Type Analysis showing the number of donors across different blood types.

Worksheet 3: Donation Trend Over Time

Visualization: Line Chart

Purpose:

To analyze the trend in the number of donors over time based on their last recorded donation date.

Fields Used:

- **Last Donation Date** — Columns, grouped by Month
- **Donor Id** — Rows
- **Aggregation:** Count Distinct (CNTD(Donor Id))
- **Mark Type:** Line

Insights

1. The number of donors with recorded last-donation dates **generally increases over the observed period.**
2. The chart shows a **marked increase in recorded donors around 2024**, reaching values above 1,000 in some months.

3. Donation activity is relatively low in the earlier years and becomes considerably higher in the later period.

The monthly trend shows an overall increase in the number of donors based on their last recorded donation date. The later period, particularly around 2024, shows substantially higher recorded donor counts compared with the earlier years.

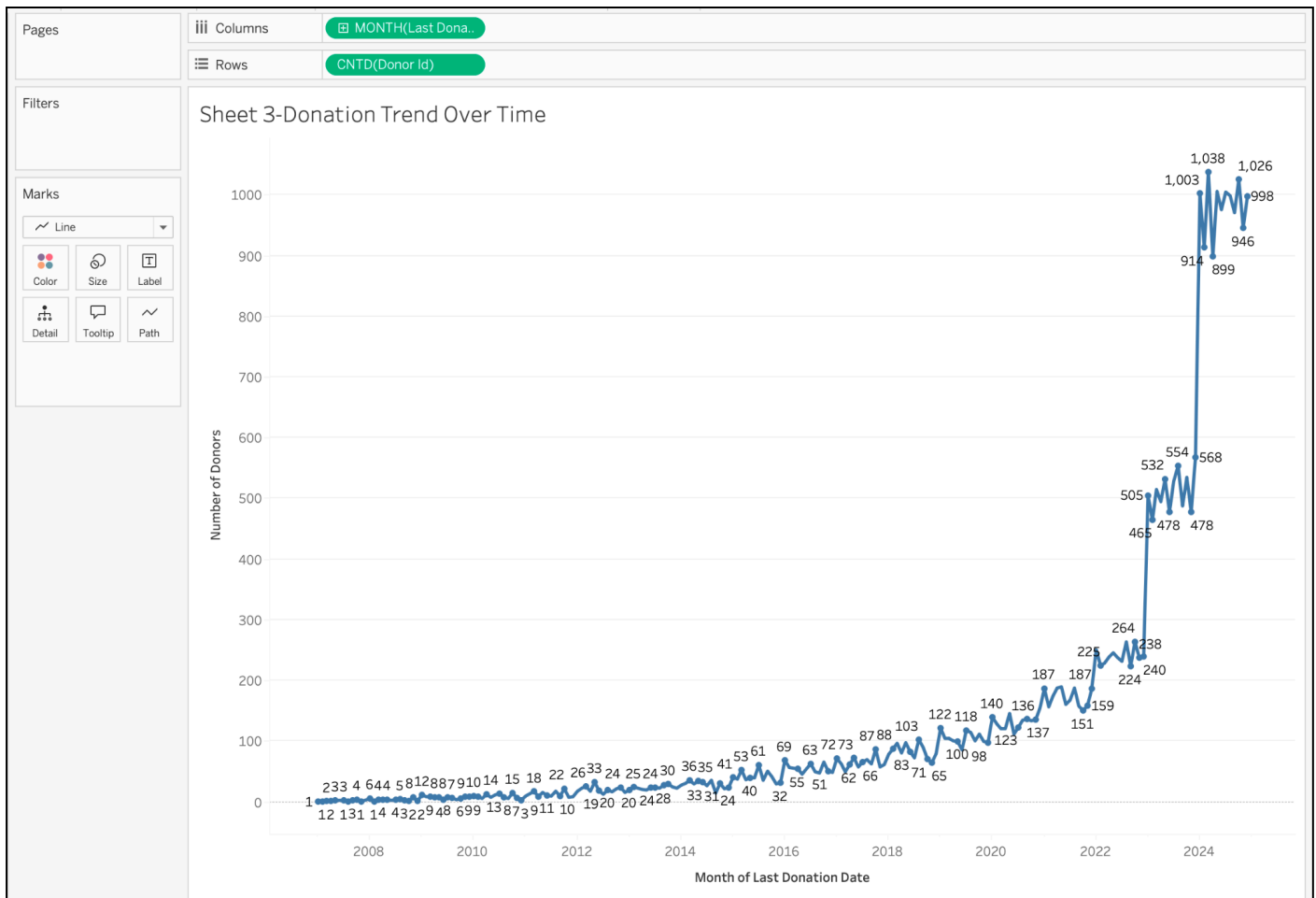


Figure 3: Donation Trend Over Time showing the monthly distribution of donors based on their last recorded donation date.

Worksheet 4: Age Group Analysis

Visualization: Bar Chart

Purpose:

To analyze the distribution of donors across different age groups and identify the age groups with the highest and lowest number of donors.

Fields Used:

- **Age Group** — Columns
- **Donor Id** — Rows
- **Aggregation:** Count Distinct (CNTD(Donor Id))
- **Mark Type:** Bar
- **Mark Labels:** Enabled

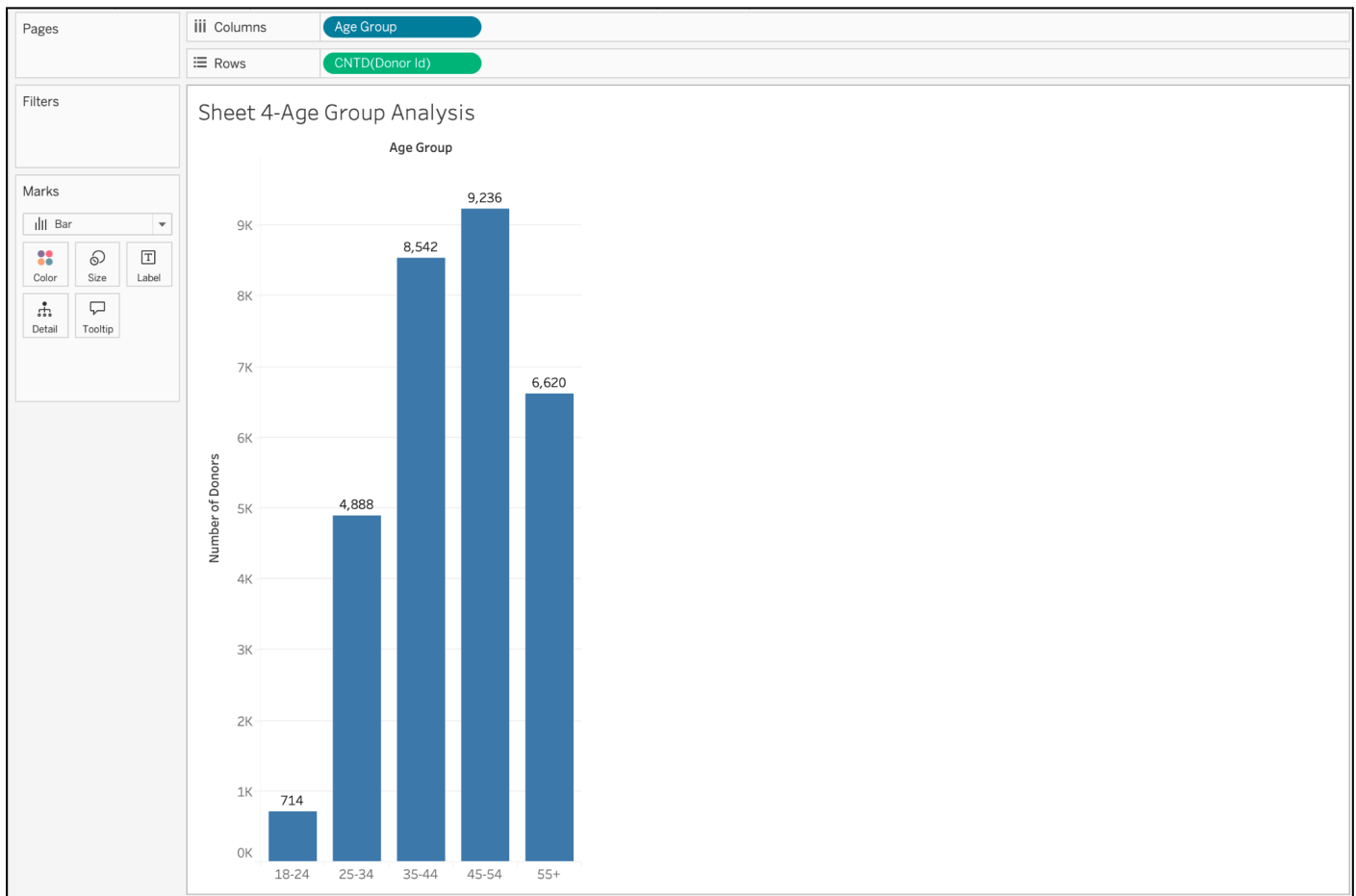


Figure 4: Age Group Analysis showing the distribution of donors across different age groups.

INSIGHT:

1. The **45–54 age group** has the **highest number of donors**, with **9,236 donors**.
2. The **18–24 age group** has the **lowest number of donors**, with **714 donors**.
3. The **35–44 age group** is the second-largest group, with **8,542 donors**.
4. The **55+ age group** also represents a substantial portion of the donor dataset, with **6,620 donors**.

The **45–54 age group** has the highest number of donors, with **9,236 donors**, while the **18–24 age group** has the lowest, with **714 donors**. The **35–44** and **55+** age groups also show relatively high donor counts compared with the younger age groups.

Sheet 5: Donor Distribution Map

Purpose:

To visualize the geographical distribution of donors across different countries and identify regions with higher concentrations of donors.

Fields Used:

- Country Code → Detail
- Latitude (Generated) → Rows
- Longitude (Generated) → Columns
- Donor Id → Count Distinct
- Count Distinct Donor Id → Size

- Mark Type → Circle

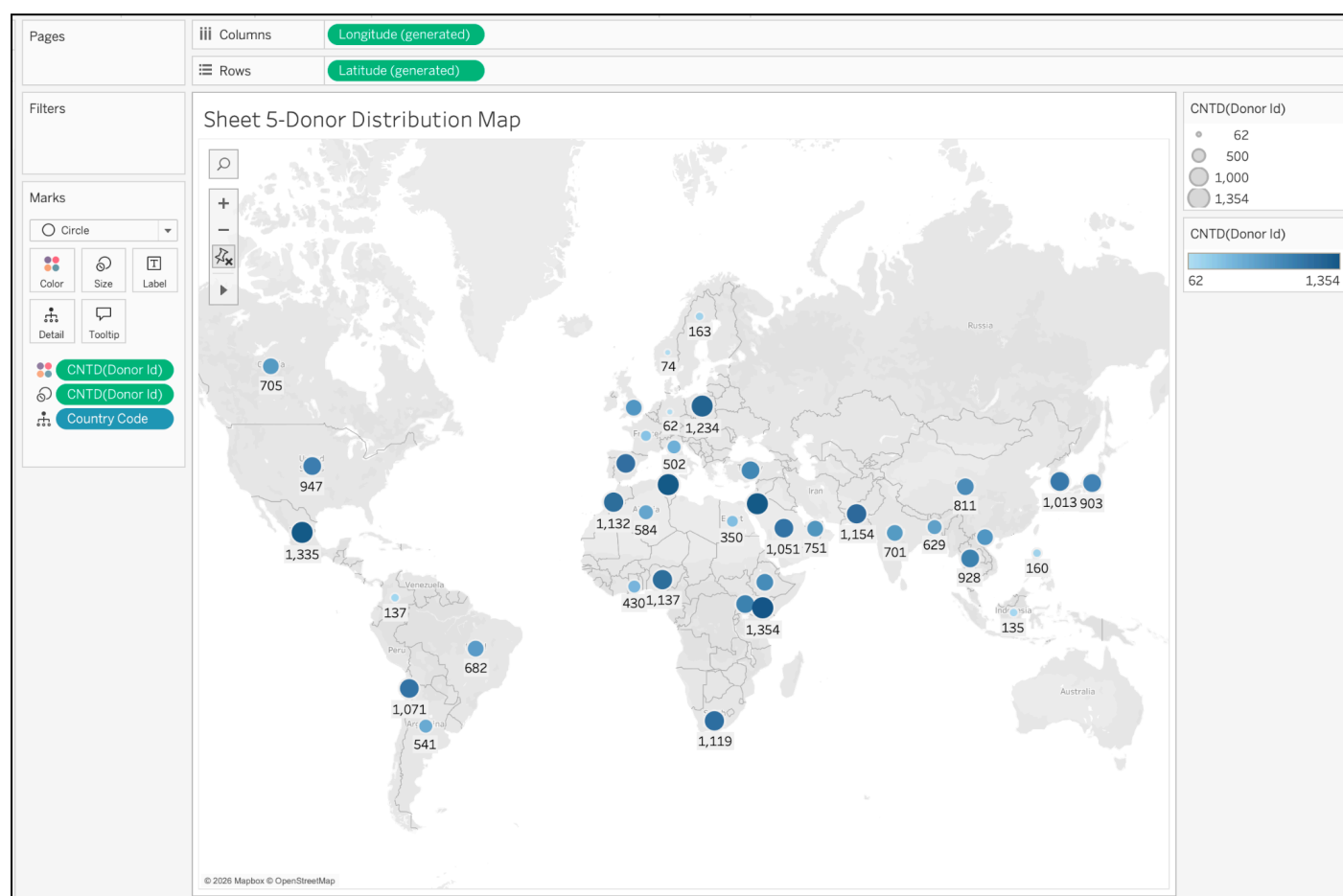


Figure 5: Donor Distribution Map showing the geographical distribution and concentration of donors across countries.

INSIGHT:

1. Donors are distributed across multiple countries and regions.
2. The size of the circles varies according to the number of recorded donors.
3. Some countries show noticeably larger donor concentrations than others.

The geographical distribution map shows that the donor population is spread across multiple countries, with varying donor concentrations between locations. Larger bubbles indicate countries with higher recorded donor counts, helping identify areas with relatively greater donor representation.

Worksheet 6: Eligibility Status Analysis

Visualization: Bar Chart

Purpose:

To analyze the eligibility status of donors and compare the number of eligible, permanently deferred, and temporarily deferred donors.

Fields Used:

- **Eligibility Status** — Columns
- **Donor Id** — Rows
- **Aggregation:** Count Distinct (CNTD(Donor Id))

- **Mark Type:** Bar
- **Mark Labels:** Enabled

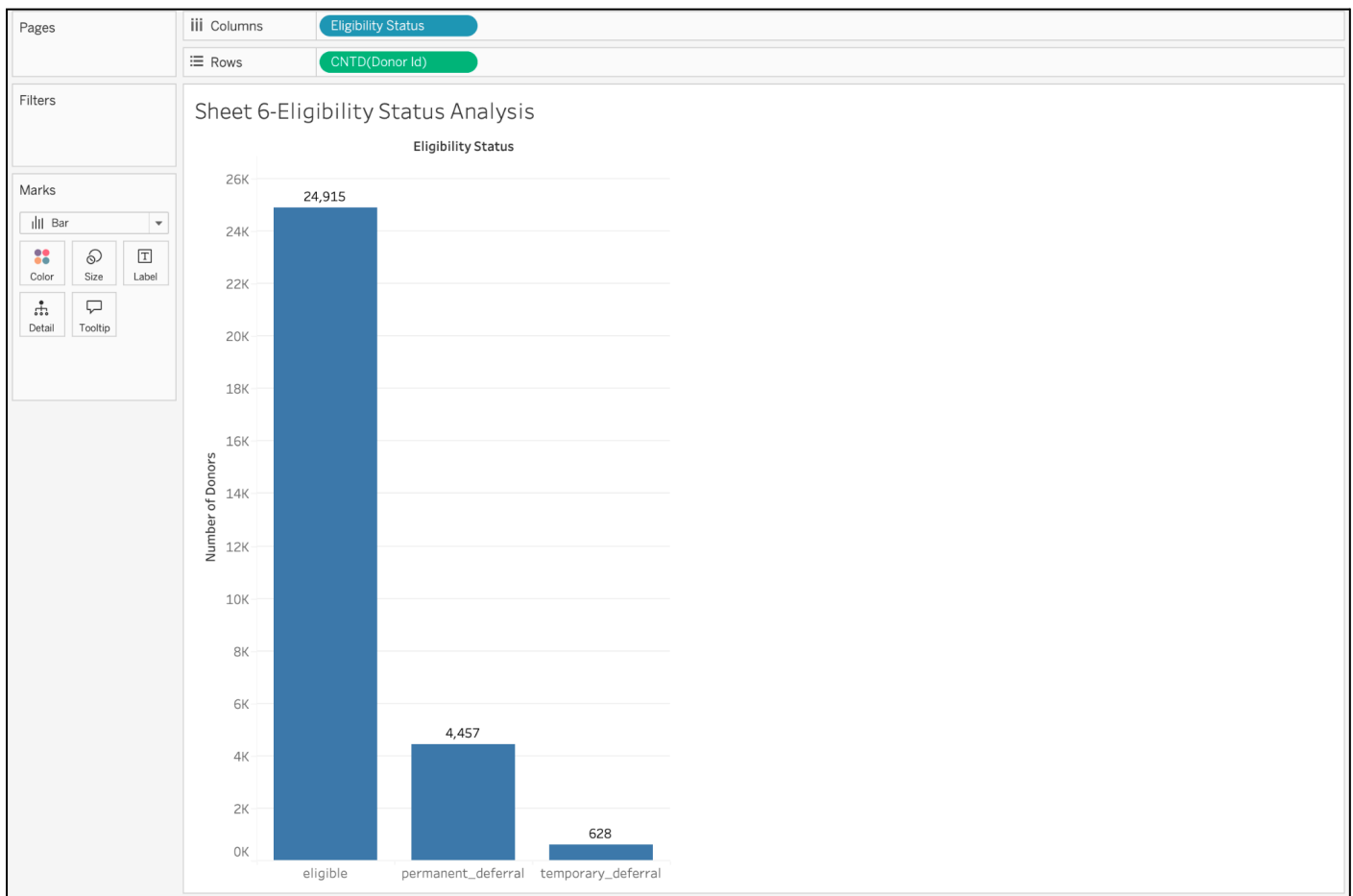


Figure 6: Eligibility Status Analysis showing the distribution of eligible, permanently deferred, and temporarily deferred donors.

INSIGHT:

1. **Eligible donors form the largest category**, with **24,915 donors**.
2. **Permanent deferrals** account for **4,457 donors**.
3. **Temporary deferrals** account for **628 donors**, making it the smallest category.
4. The three eligibility categories together account for **all 30,000 donors** in the dataset.

The eligibility analysis shows that the majority of donors are classified as **eligible to donate**, while a smaller portion is permanently or temporarily deferred. Permanent deferrals represent a considerably larger group than temporary deferrals.

Worksheet 7: Deferral Reasons Analysis

Visualization: Bar Chart

Purpose:

To analyze the major reasons for donor deferral and compare the number of donors affected by each deferral reason.

Fields Used:

- **Deferral Reason** — Columns
- **Donor Id** — Rows

- **Aggregation:** Count Distinct (CNTD(Donor Id))
- **Filter:** Null values excluded
- **Mark Type:** Bar
- **Mark Labels:** Enabled

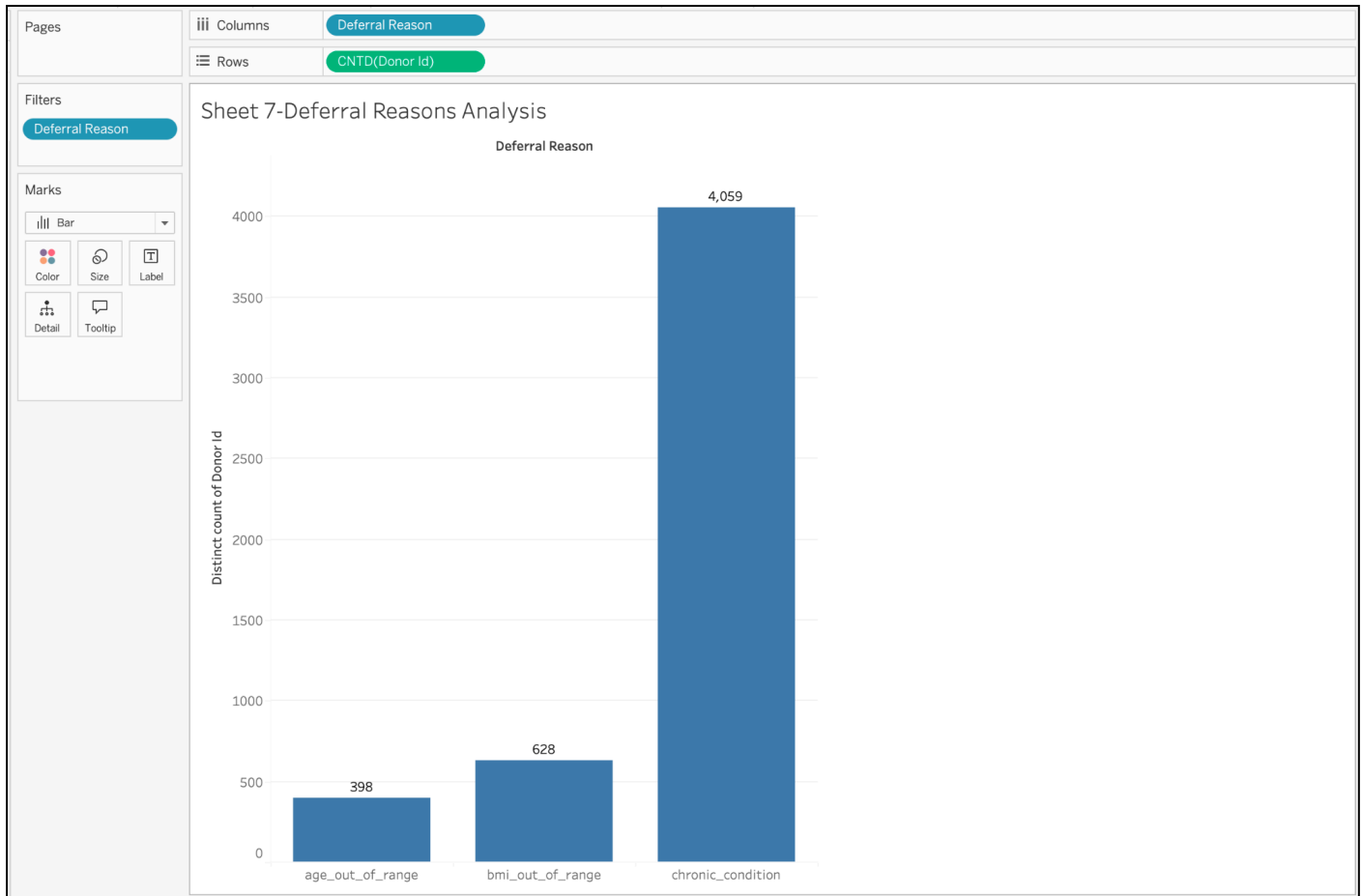


Figure 7: Deferral Reasons Analysis showing the distribution of donors across different deferral reasons.

INSIGHT:

1. **Chronic condition is the most common deferral reason**, with **4,059** donors.
2. **BMI out of range** is the second most common reason, affecting **628** donors.
3. **Age out of range** is the least common recorded deferral reason, affecting **398** donors.
4. The three recorded deferral reasons account for **5,085** deferred donors in total.

The deferral analysis shows that **chronic conditions are the dominant recorded reason for donor deferral**, while BMI-related and age-related reasons account for smaller portions of the deferred donor population.

Worksheet 8: Donation Frequency Analysis

Visualization: Bar Chart

Purpose:

To analyze the distribution of donors based on the number of donations recorded during the last 12 months.

Fields Used:

- **Donation Frequency Group** — Columns

- **Donor Id** — Rows
- **Aggregation:** Count Distinct (CNTD(Donor Id))
- **Mark Type:** Bar
- **Mark Labels:** Enabled

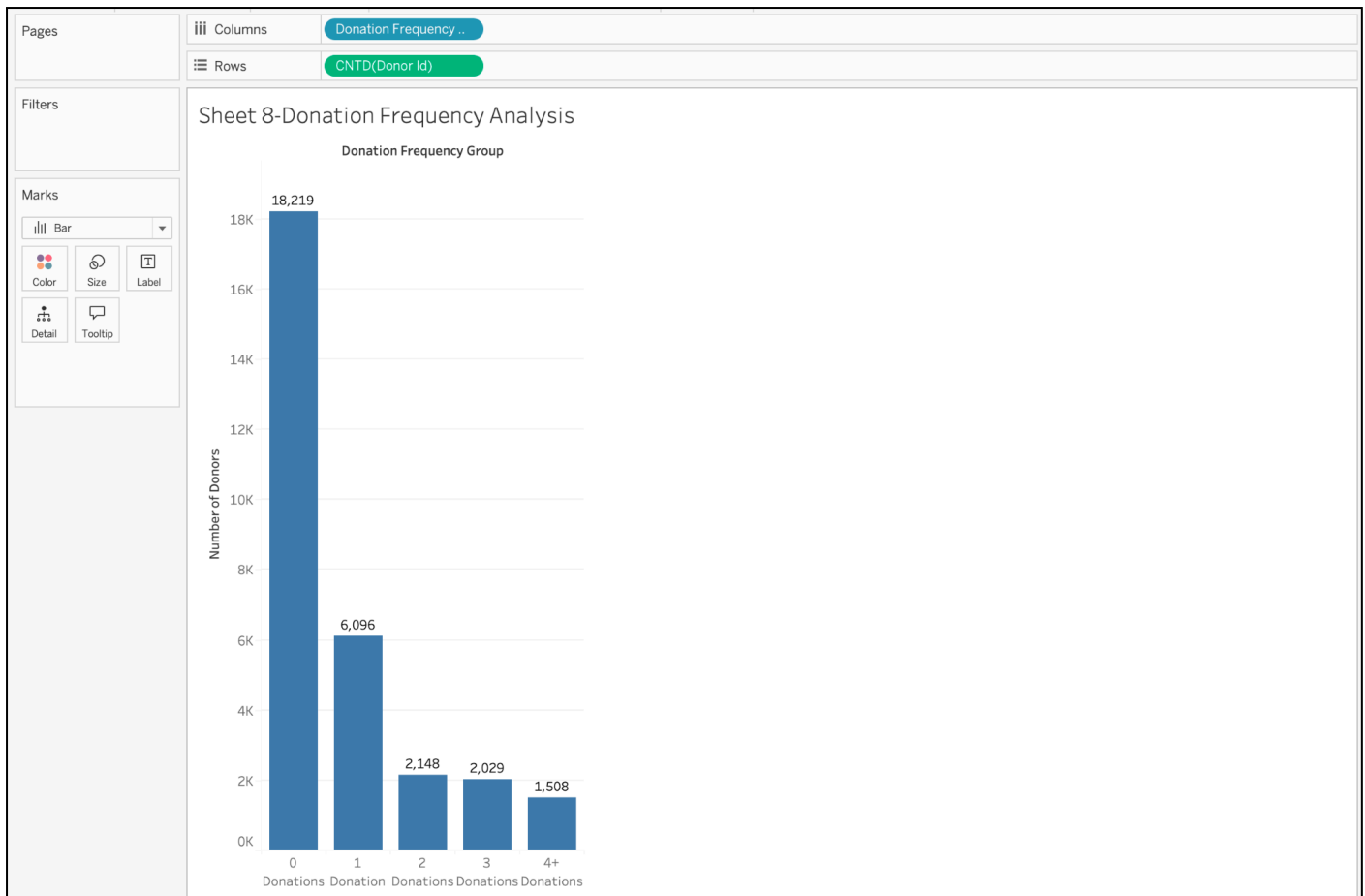


Figure 8: Donation Frequency Analysis showing the distribution of donors according to their donation frequency during the last 12 months.

INSIGHT:

1. The **0 Donations** group has the highest number of donors, with **18,219 donors**.
2. **6,096 donors** recorded exactly **1 donation** during the last 12 months.
3. The **2 Donations** and **3 Donations** groups contain **2,148** and **2,029 donors**, respectively.
4. The **4+ Donations** group is the smallest, with **1,508 donors**.

The donation frequency analysis shows that a large portion of the donor population had **no recorded donation during the last 12 months**, while the number of donors generally decreases as donation frequency increases. A comparatively smaller group recorded three or more donations during the period.

Worksheet 9: Regular Donor Analysis

Visualization: Bar Chart

Purpose:

To compare the number of regular and non-regular donors in the dataset.

Fields Used:

- **Donor Type** — Columns

- **Donor Id** — Rows
- **Aggregation:** Count Distinct (CNTD(Donor Id))
- **Calculated Field:** Donor Type
- **Mark Type:** Bar
- **Mark Labels:** Enabled

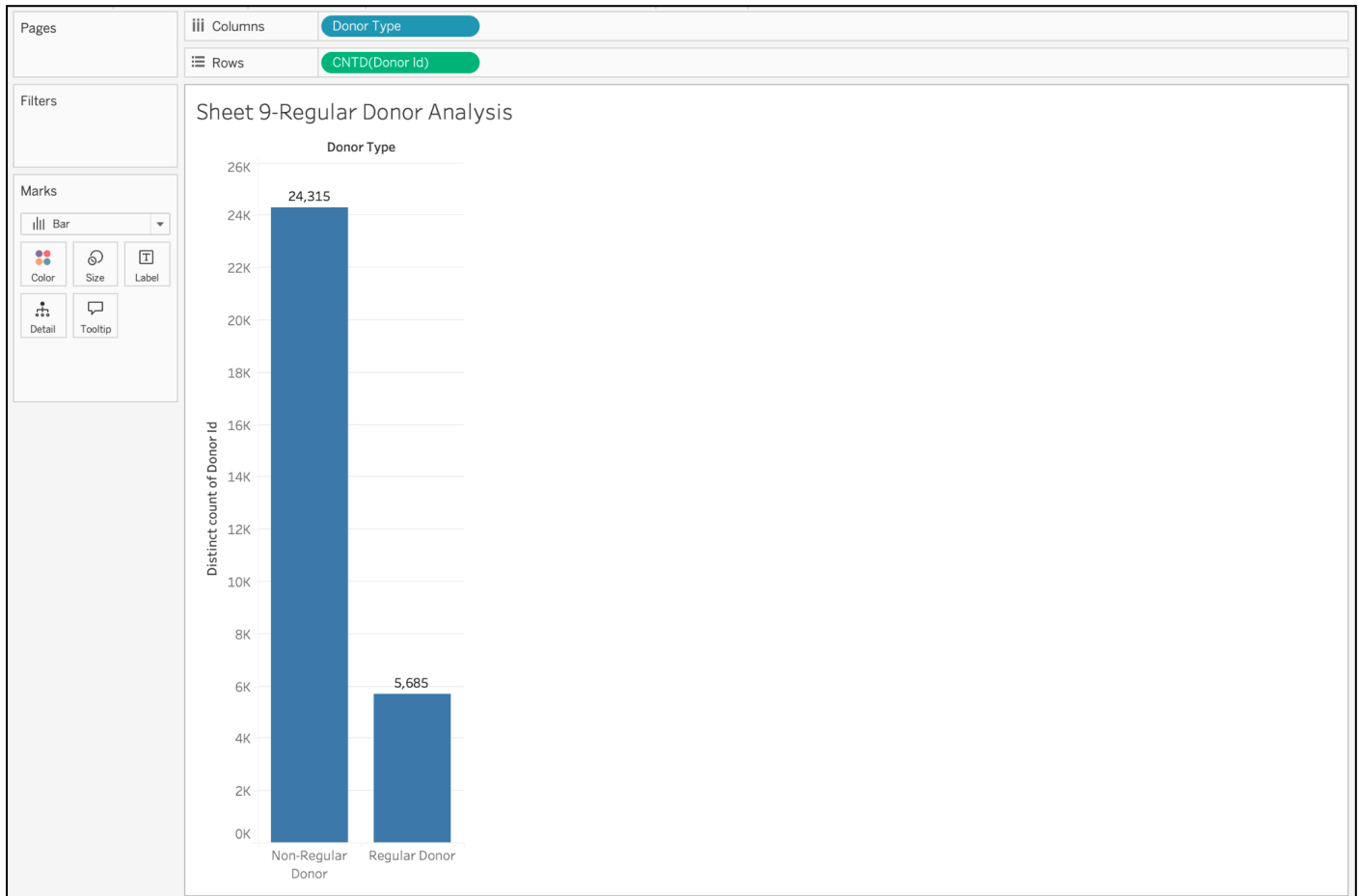


Figure 9: Regular Donor Analysis showing the distribution of regular and non-regular donors.

INSIGHT:

1. **Non-Regular Donors form the larger group, with 24,315 donors.**
2. **Regular Donors account for 5,685 donors in the dataset.**
3. The two categories together represent **all 30,000 donors**.
4. Regular donors represent a smaller portion of the overall donor population compared with non-regular donors.

The regular donor analysis shows that the dataset contains a substantially larger number of **non-regular donors** than regular donors. Out of 30,000 donors, **5,685 are classified as regular donors**, while 24,315 are classified as non-regular donors.

Sheet 10 — Donor Segmentation Using Frequency and Recency

Visualization: Scatter Plot with Clustering

Purpose:

To segment donors into groups based on their donation frequency during the last 12 months and the number of days since their last donation.

Fields Used:

- Donation Count Last 12M → Columns
- Recency Days → Rows
- Donor Id → Detail
- Clusters → Color
- Mark Type → Circle
- Number of Clusters → Automatic

Clustering Variables:

- Donation Count Last 12M
- Recency Days

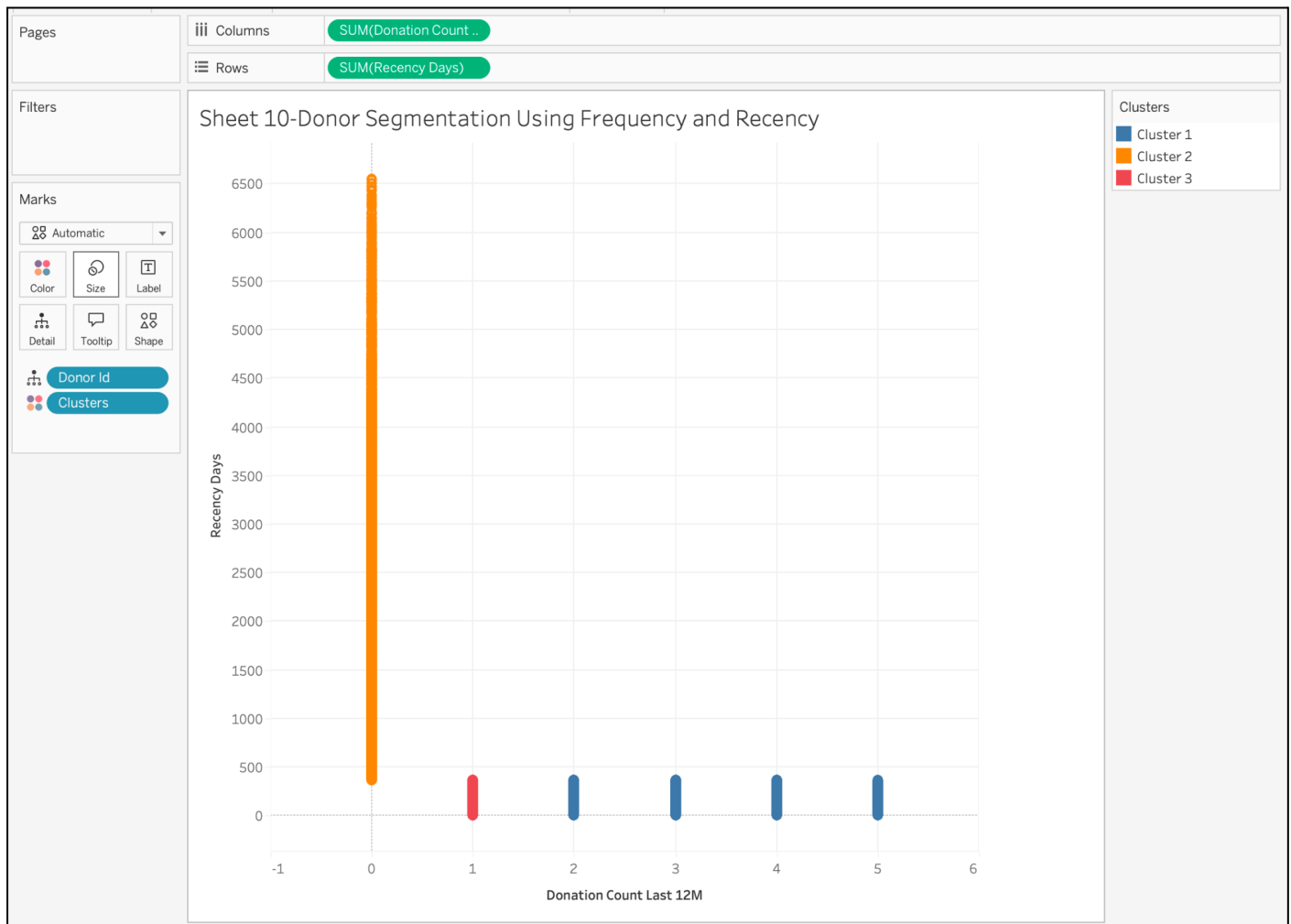


Figure 10: Donor Segmentation using Frequency and Recency with Tableau clustering.

INSIGHT:

1. Donors are divided into **three distinct clusters** based on donation frequency and recency.
2. One cluster contains donors with **higher recent donation activity**.
3. Other clusters represent donors with **lower donation frequency or different recency patterns**.

The clustering analysis segments donors according to their recent donation frequency and recency, helping identify different donor behavior patterns and distinguish more active donors from less active donors.

7. Calculated Fields:

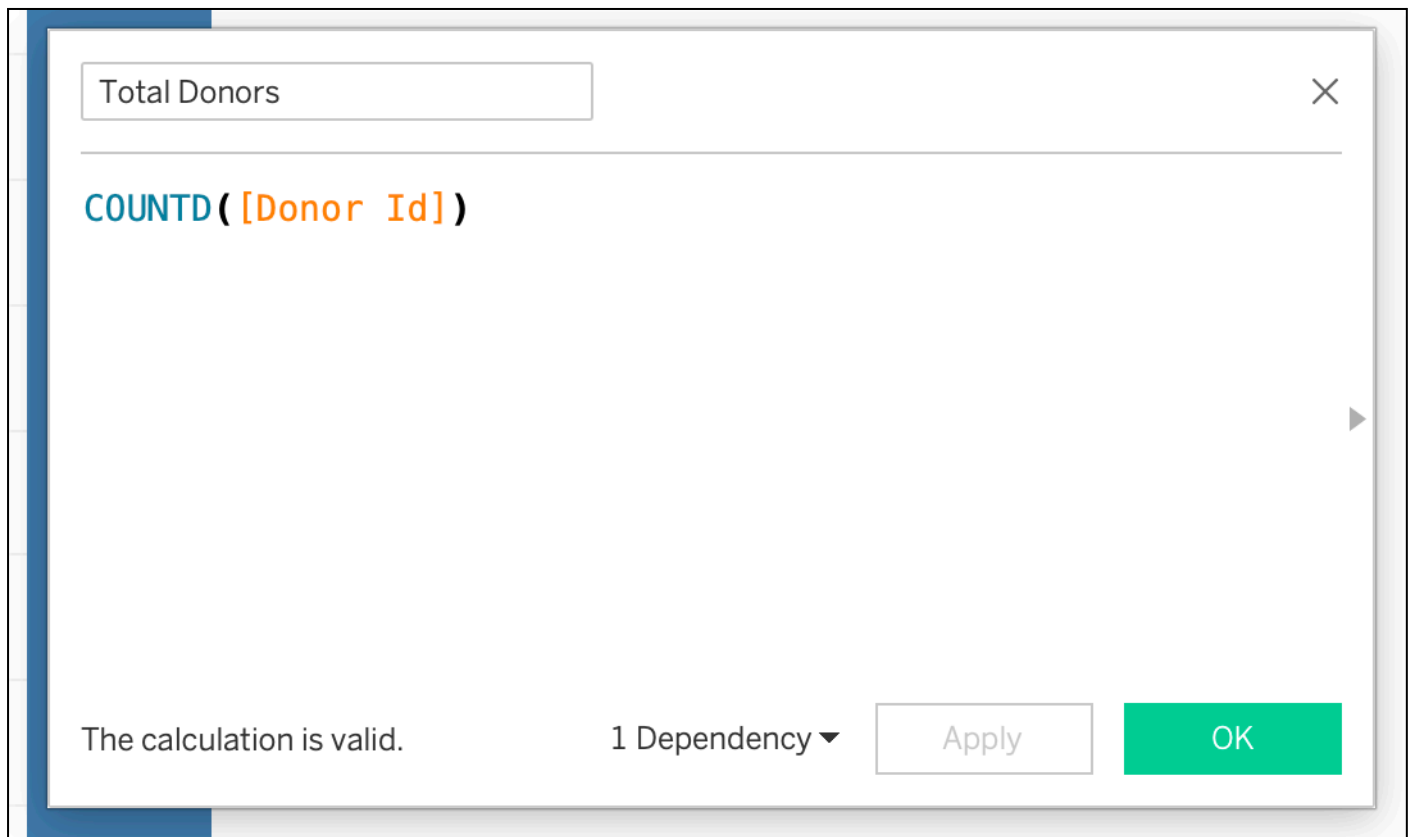
The following calculated fields were created in Tableau to transform the raw donor data into meaningful categories and metrics for dashboard analysis.

1. Total Donors

Purpose: To calculate the total number of unique donors in the dataset.

CODE: COUNTD([Donor Id])

Used in: KPI Cards



2. Age Group

Purpose: To categorize donors into meaningful age groups for demographic analysis.

CODE:

```
IF [Age] < 25 THEN "18-24"
ELSEIF [Age] < 35 THEN "25-34"
ELSEIF [Age] < 45 THEN "35-44"
ELSEIF [Age] < 55 THEN "45-54"
ELSE "55+"
END
```

Used in: Age Group Analysis

Age Group

```
IF [Age] < 25 THEN "18-24"
ELSEIF [Age] < 35 THEN "25-34"
ELSEIF [Age] < 45 THEN "35-44"
ELSEIF [Age] < 55 THEN "45-54"
ELSE "55+"
END
```

The calculation is valid. 1 Dependency Apply OK

3. Donation Frequency Group

Purpose: To classify donors according to the number of donations recorded during the last 12 months.

CODE:

```
IF [Donation Count Last 12M] = 0 THEN "0 Donations"
ELSEIF [Donation Count Last 12M] = 1 THEN "1 Donation"
ELSEIF [Donation Count Last 12M] = 2 THEN "2 Donations"
ELSEIF [Donation Count Last 12M] = 3 THEN "3 Donations"
ELSE "4+ Donations"
END
```

Used in: Donation Frequency Analysis

Donation Frequency Group

```
IF [Donation Count Last 12M] = 0 THEN "0 Donations"
ELSEIF [Donation Count Last 12M] = 1 THEN "1 Donation"
ELSEIF [Donation Count Last 12M] = 2 THEN "2 Donations"
ELSEIF [Donation Count Last 12M] = 3 THEN "3 Donations"
ELSE "4+ Donations"
END
```

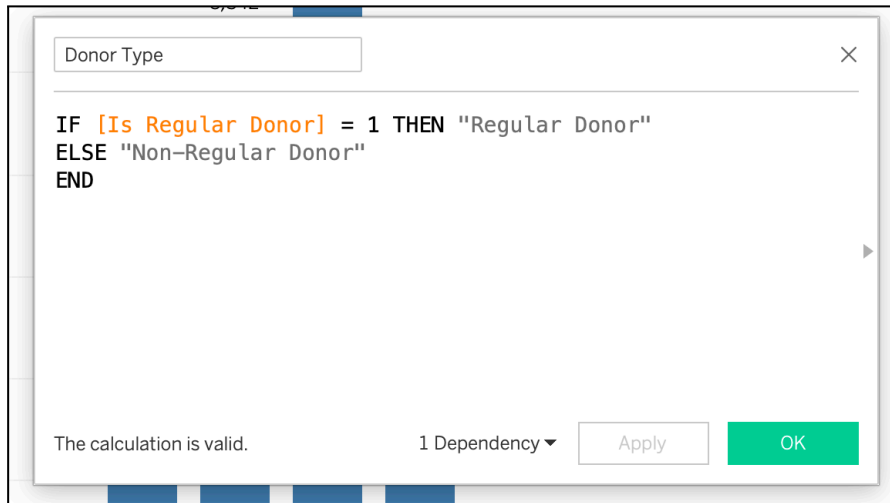
The calculation is valid. 1 Dependency Apply OK

4. Donor Type

Purpose: To classify donors as regular or non-regular donors.

CODE:

```
IF [Is Regular Donor] = 1 THEN "Regular Donor"
ELSE "Non-Regular Donor"
END
```



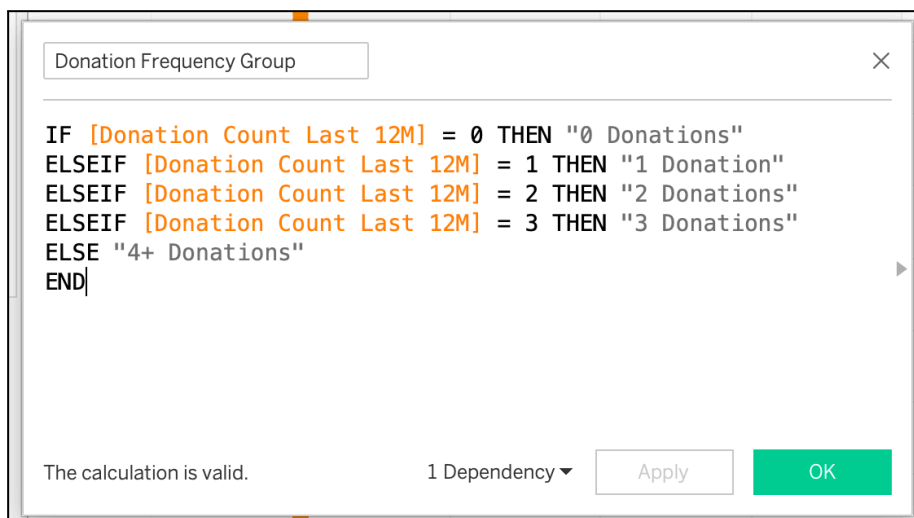
Used in: Regular Donor Analysis

5. Donation Frequency Group

Purpose: To classify donors into groups based on the number of donations recorded during the last 12 months.

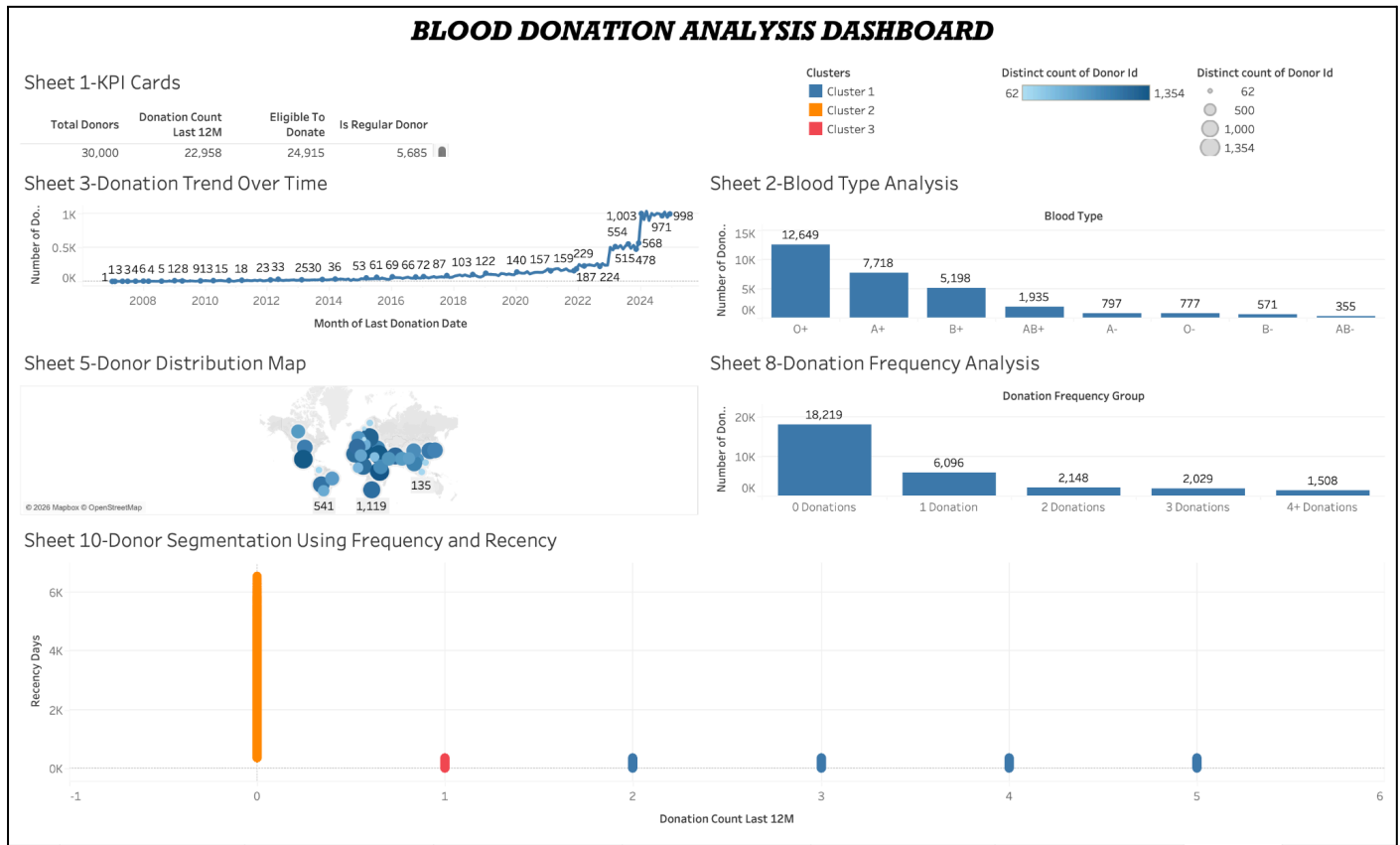
CODE:

```
IF [Donation Count Last 12M] = 0 THEN "0 Donations"
ELSEIF [Donation Count Last 12M] = 1 THEN "1 Donation"
ELSEIF [Donation Count Last 12M] = 2 THEN "2 Donations"
ELSEIF [Donation Count Last 12M] = 3 THEN "3 Donations"
ELSE "4+ Donations"
END
```



Used in: Donation Frequency Analysis

8. Dashboard Screenshot:



9. Interactions:

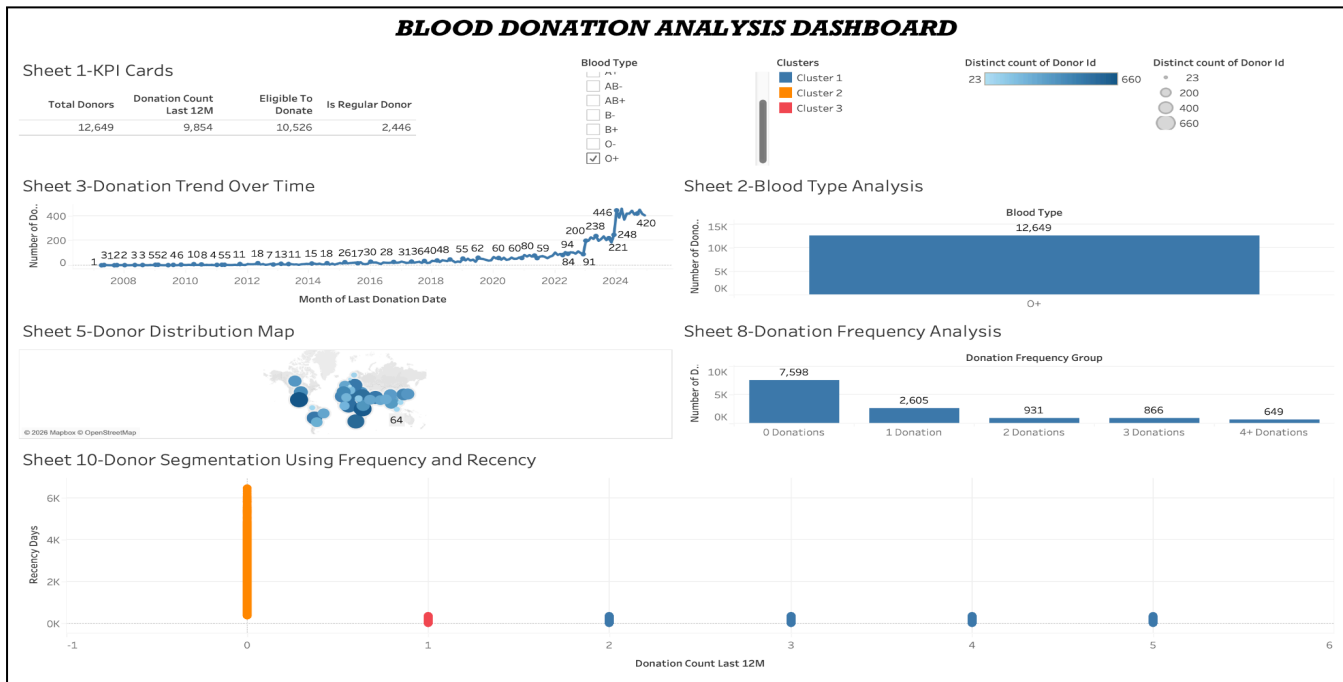
1. Blood Type Filter

Purpose: To allow users to filter the dashboard based on the selected blood type.

Interaction: A Blood Type filter was added to the dashboard, allowing users to select specific blood groups such as A+, B+, AB+, O+, and their negative variants.

Function: When a blood type is selected, the dashboard updates to display the corresponding donor information.

Used in: Interactive Blood Donation Analysis Dashboard



Blood Type Filter shown on the dashboard

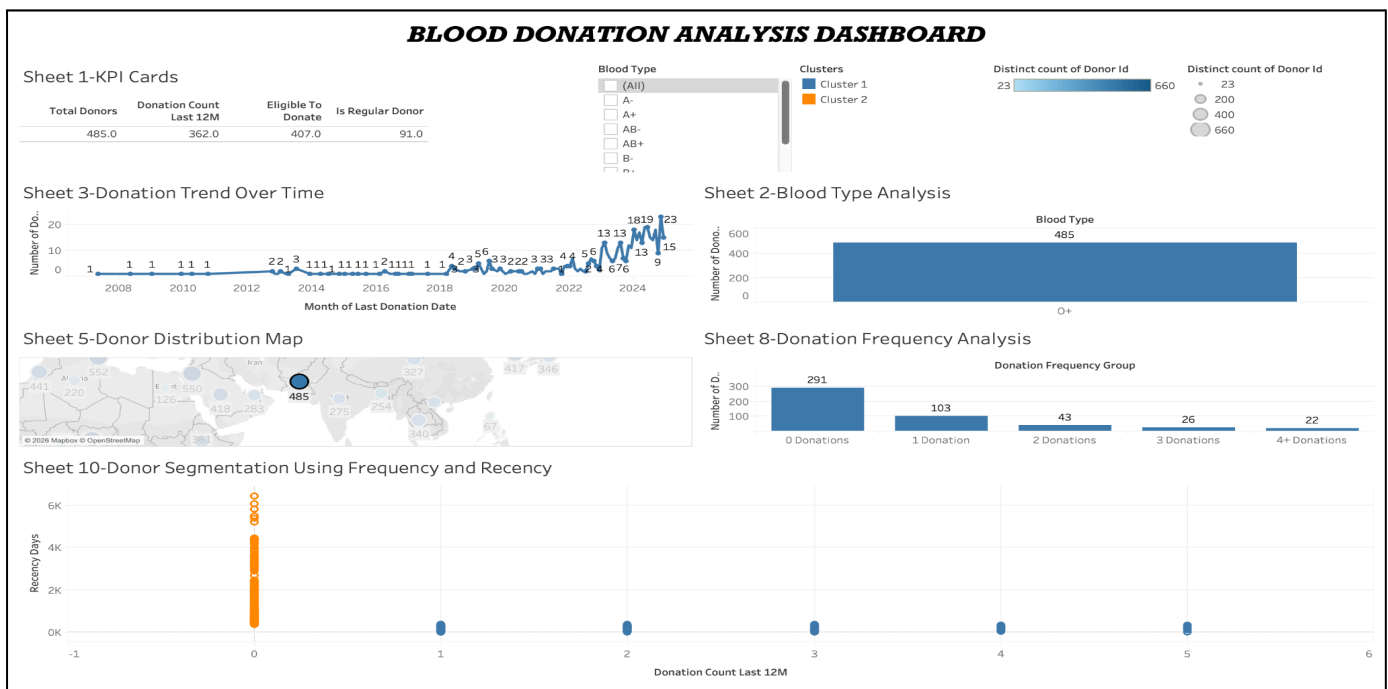
2. Region Filter

Purpose: To allow users to filter the dashboard based on the selected geographical region.

Interaction: A Region filter was added to the dashboard, allowing users to select different regions such as Asia, Africa, Europe, Americas, and MENA.

Function: When a region is selected, the dashboard updates to display the corresponding donor information for that region.

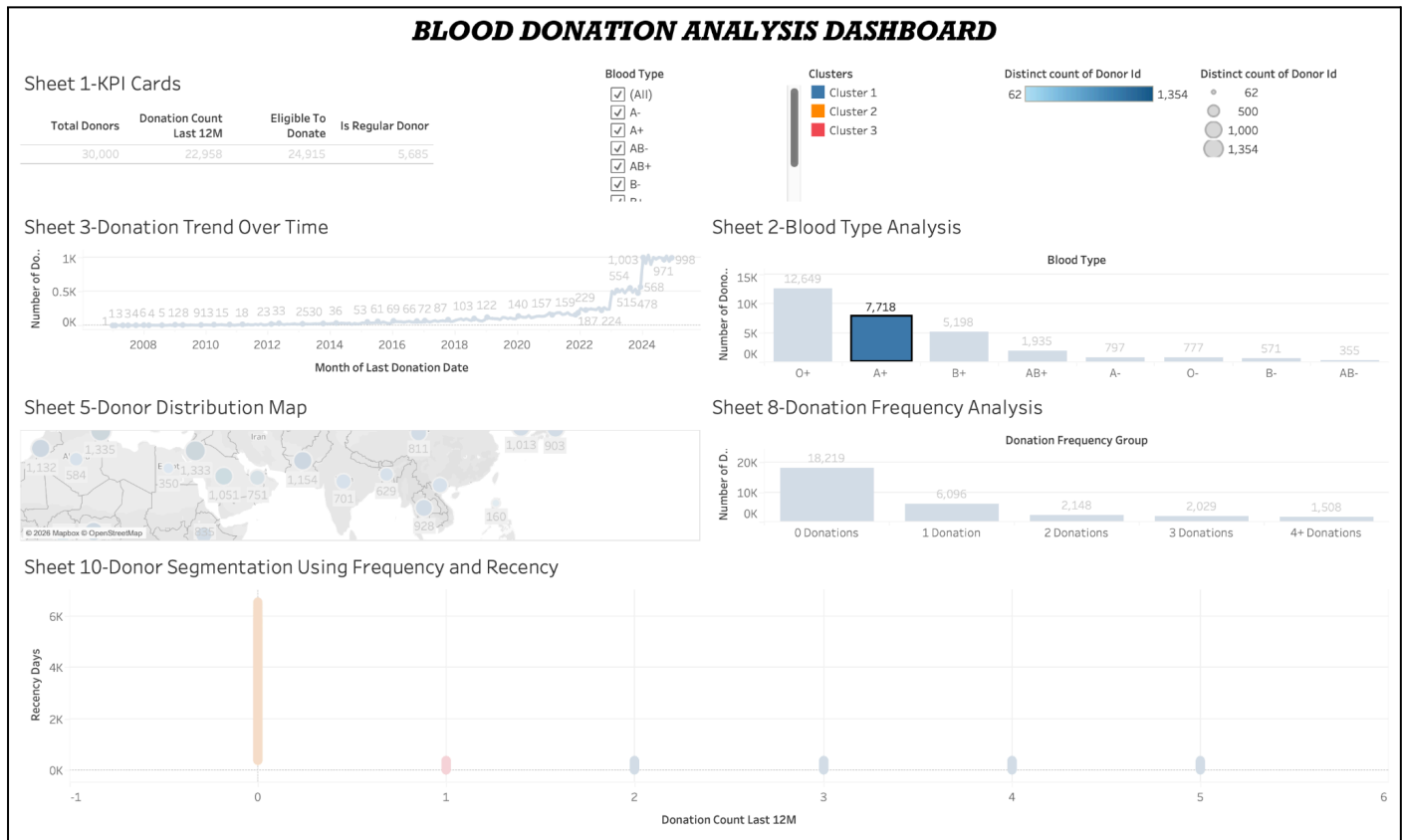
Used in: Interactive Blood Donation Analysis Dashboard



Region Filter shown on the dashboard

dashboard visualizations remain visible, allowing users to compare the selected blood type with the remaining data.

Used in: Interactive Blood Donation Analysis Dashboard



Blood Type Analysis with selected blood type highlighted

5. Cluster Selection / Filter Action

Purpose:

To allow users to explore donor segments by selecting a specific cluster from the Donor Segmentation visualization.

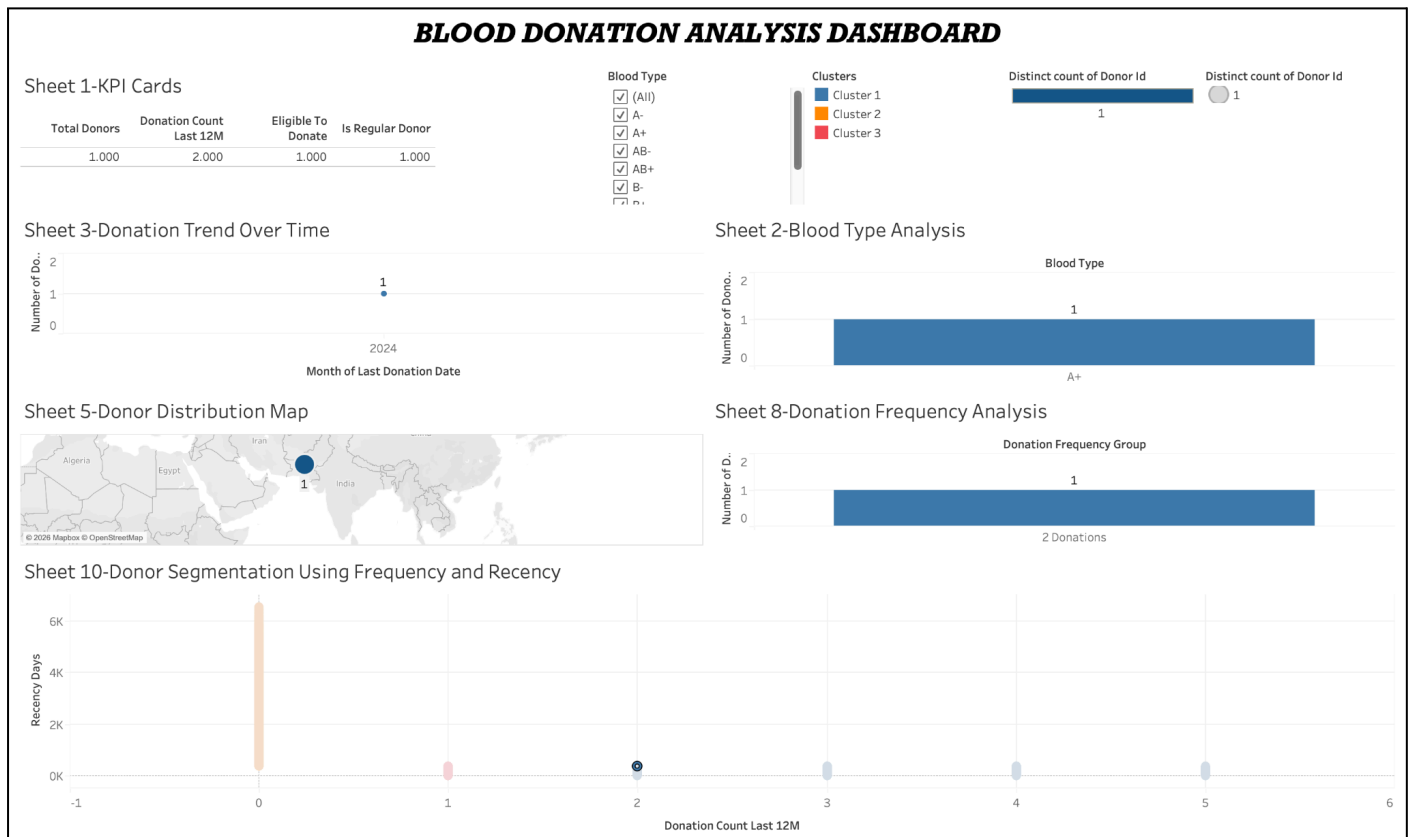
Interaction:

A filter action was added to the **Donor Segmentation Using Frequency and Recency** chart. When a cluster is selected, the dashboard filters the other visualizations according to the selected donor segment.

Function:

When a cluster is selected, the **Blood Type Analysis**, **Donation Trend**, **Donor Distribution Map**, and **Donation Frequency Analysis** update to display information for the selected cluster. Clearing the selection restores all values.

Used in: Interactive Blood Donation Analysis Dashboard



Donor Segmentation chart with a selected cluster and filtered dashboard

6. Interactive Tooltip

Purpose:

To provide detailed information when the user hovers over a visualization.

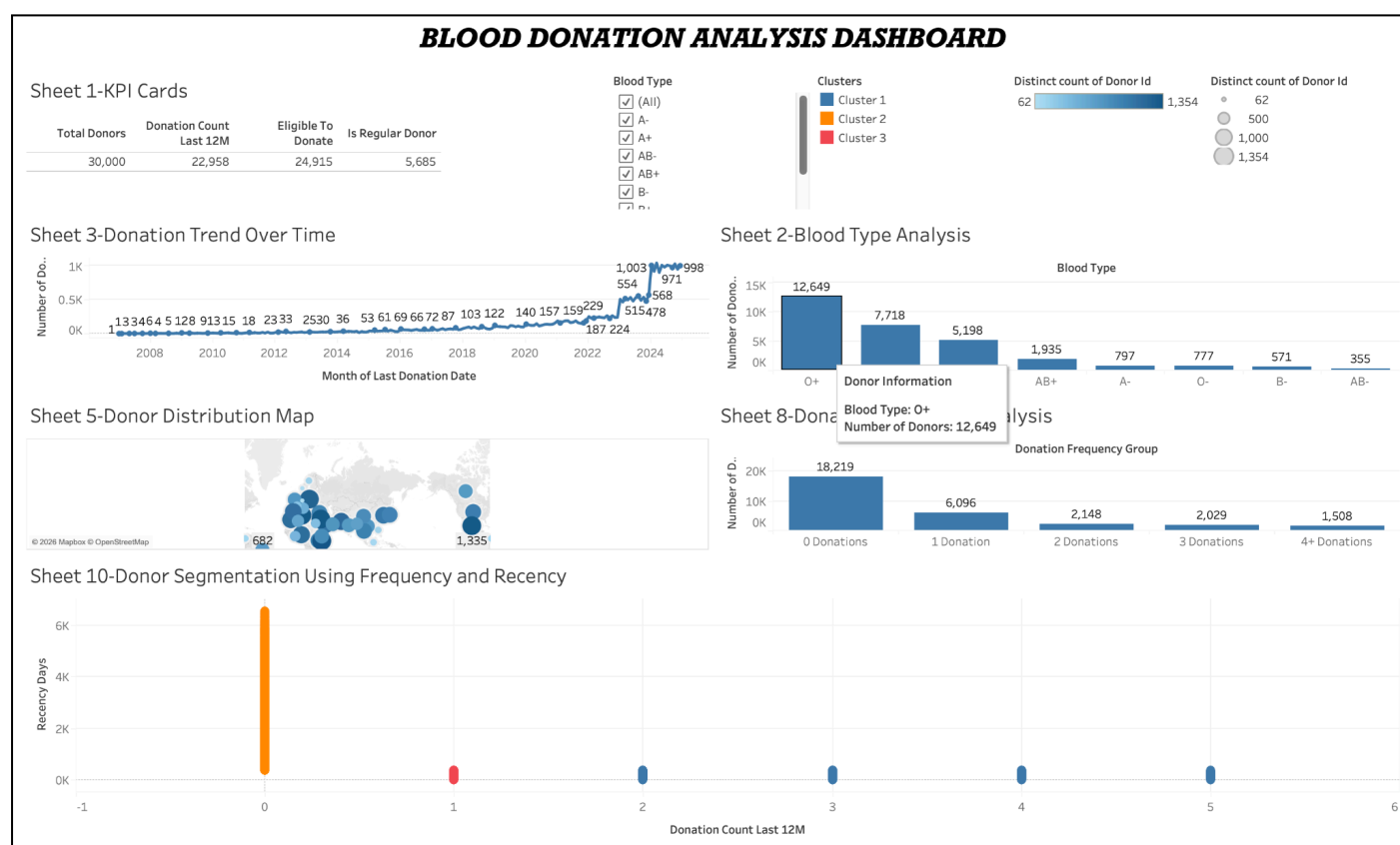
Interaction:

An interactive tooltip was customized for the **Blood Type Analysis** chart. When the user hovers over a blood type bar, the tooltip displays the selected blood type and the corresponding number of donors.

Function:

The tooltip provides additional information without changing or filtering the dashboard. It displays the **Blood Type** and **Number of Donors** for the selected bar.

Used in: Interactive Blood Donation Analysis Dashboard



Interactive Tooltip displaying blood type and number of donors

10. Key Insights

- The dataset contains 30,000 unique donors, with 24,915 eligible donors and 5,685 regular donors, as shown through the KPI cards.**
- O+ is the most represented blood type with 12,649 donors, while AB- has the lowest representation with 355 donors.**
- The 45–54 age group has the highest number of donors with 9,236, while the 18–24 age group has the lowest with 714 donors.**
- 18,219 donors have zero recorded donations during the last 12 months, while 6,096 donors have one donation and 1,508 donors have four or more donations.**
- The donor distribution map shows that donors are distributed across multiple countries and regions, with different geographical concentrations.**

11. Video Demonstration:

Video Link: <https://youtu.be/e0uoh3z0Lm4>

Description: The demonstration video presents the Blood Donation Analysis Dashboard and shows the major dashboard components and interactive features, including blood type filtering, region filtering, map selection, highlight action, cluster selection, and interactive tooltips.

YouTube Visibility: Unlisted

12. GitHub Repository:

GitHub Link:

<https://github.com/himani-pixel/Tableau-Practical---T016/tree/main/Project#:~:text=Practical%204-,Project,-Dashboard.png>

Recommended Files to Upload

- Tableau Workbook (**.twb** / **.twbx**)
- Dataset (**.csv**)
- Final Dashboard Screenshot (**.png**)
- README file
- Project Documentation, if required

The original submission template also recommends the Tableau workbook, dataset, dashboard screenshot, and README file for the repository. Tableau Project Submission